Matplotlib

In [2]:
```
import matplotlib.pyplot as plt
```

Line Chart

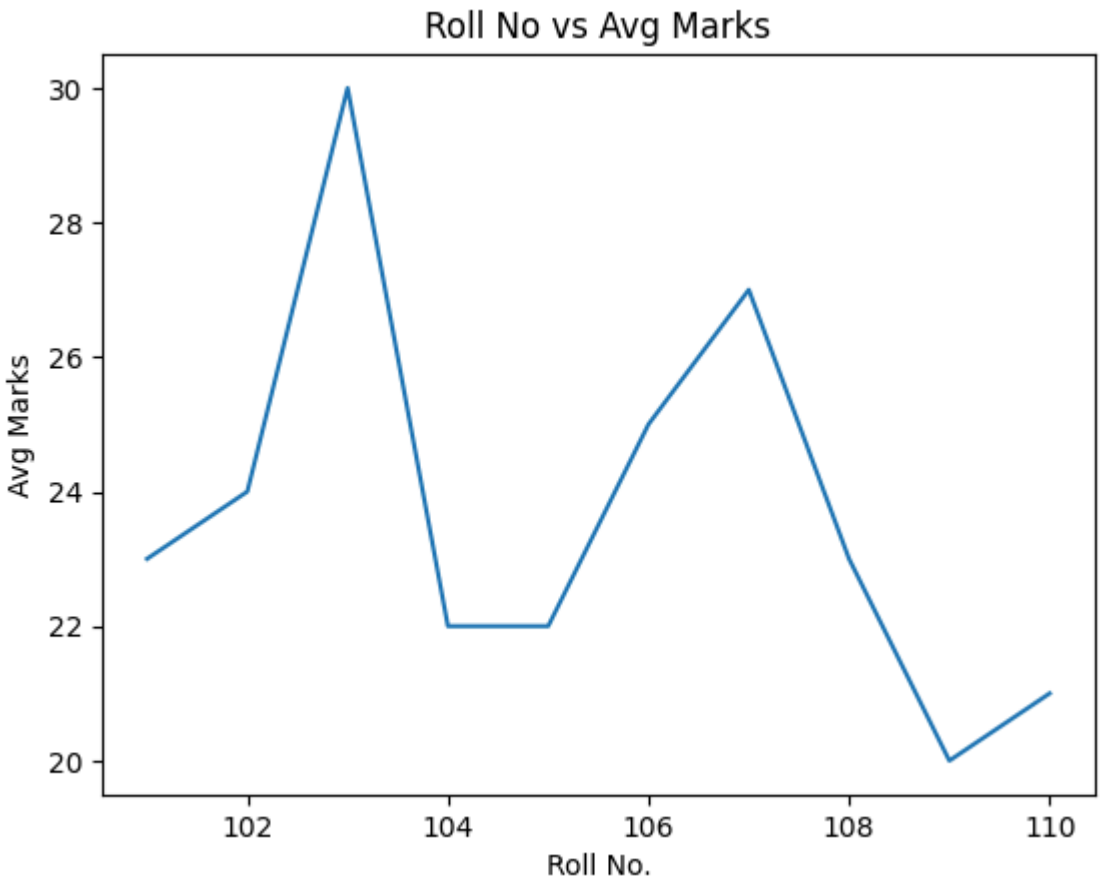
In [3]:
```
x = list(range(101,111))
```

In [6]:
```
x
```

Out[6]:
```
[101, 102, 103, 104, 105, 106, 107, 108, 109, 110]
```

In [8]:
```
y = [23, 24, 30, 22, 22, 25, 27, 23, 20, 21]
```

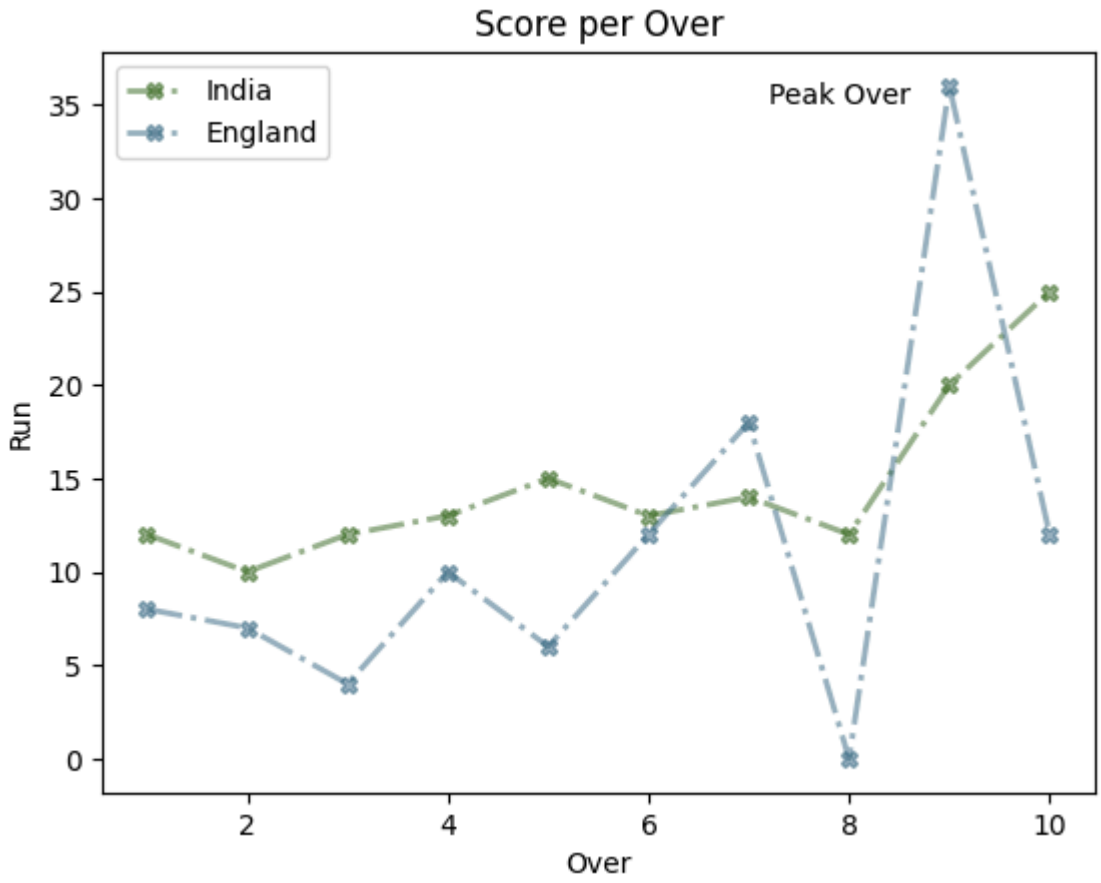
In [12]:
```
plt.plot(x, y)
plt.xlabel("Roll No.")
plt.ylabel("Avg Marks")
plt.title("Roll No vs Avg Marks")
plt.show()
```



In [14]:
```
run_ind = [12,10,12,13,15,13,14,12,20,25]
over = list(range(1,11))
```

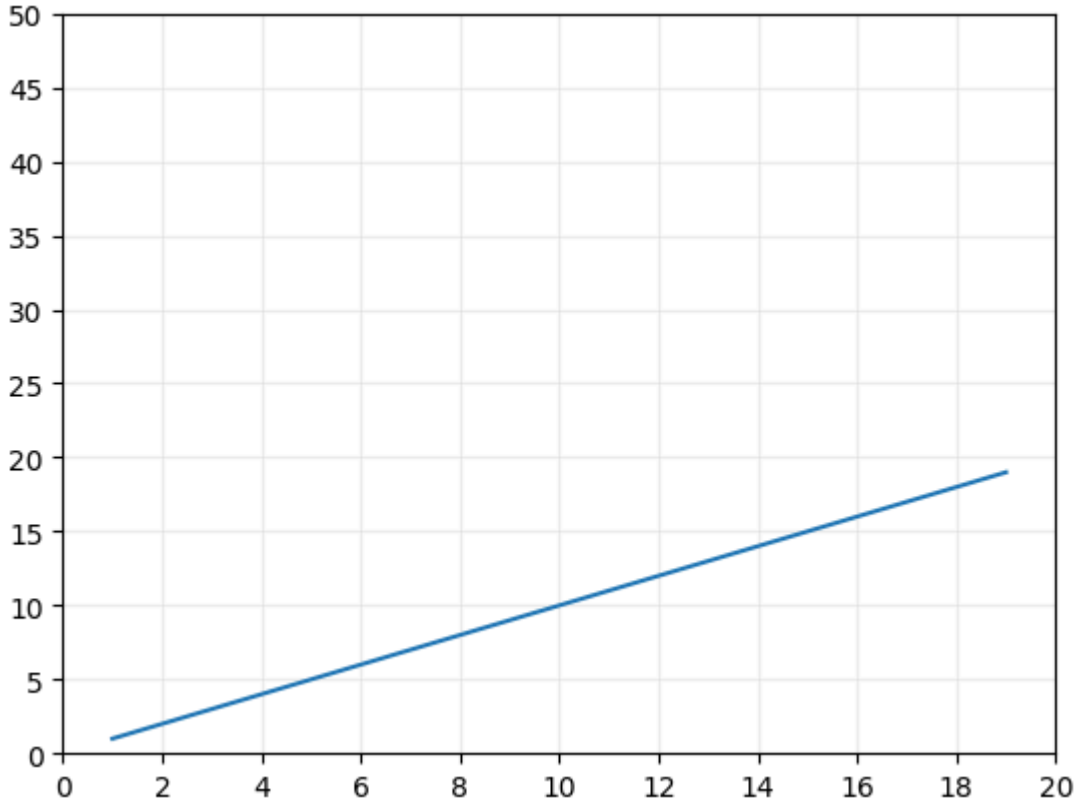
In [16]:
```
run_eng = [8,7,4,10,6,12,18,0,36,12]
```

In [18]:
```
plt.plot(over,run_ind ,marker = "X", ls = "-.", lw = "2", c = (0.2,0.4,0.1), alp
plt.plot(over,run_eng,marker = "X", ls = "-.", lw = "2", c = (0.2,0.4,0.5), alph
plt.xlabel("Over")
plt.ylabel("Run")
plt.title("Score per Over")
plt.legend(["India","England"], loc = 2)
plt.annotate("Peak Over", xy = [7.2,35])
plt.savefig("scoreboard.png", format = "png")
plt.show()
```



Custom Axis

In [22]:
```
ax = plt.axes()
ax.set_xlim([0,20])
ax.set_ylim([0,50])
ax.set_xticks(range(0,21,2))
ax.set_yticks(range(0,51,5))
ax.grid(alpha = .2)
plt.plot(range(1,21,2),range(1,21,2))
plt.show()
```



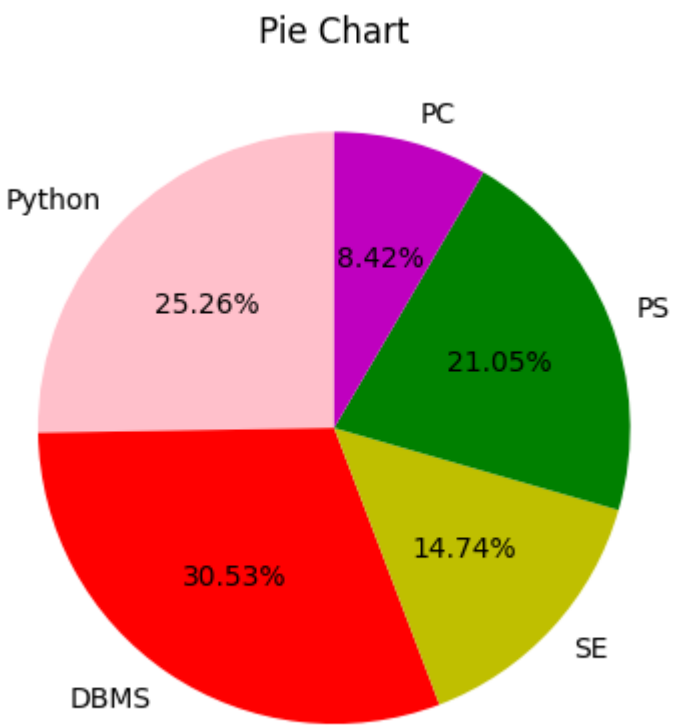
Pie Chart

In [25]:
```
subjects = ["Python","DBMS","SE","PS","PC"]
```

In [27]:
```
students = [120, 145, 70, 100, 40]
```

In [29]:
```
clr = ["pink","r","y","g","m"]
```

In [31]:
```
plt.pie(students, labels=subjects, radius = 1,autopct = "%1.2f%%", colors=clr, s
plt.title("Pie Chart")
plt.show()
```



Donut Chart

In [35]:
```
plt.pie(students)
plt.pie([100],colors=["w"], radius = 0.5)
plt.show()
```

Bar Chart

In [38]:
```
subjects = ["Python","DBMS","SE","PS","PC"]
```

In [40]:
```
import random
```

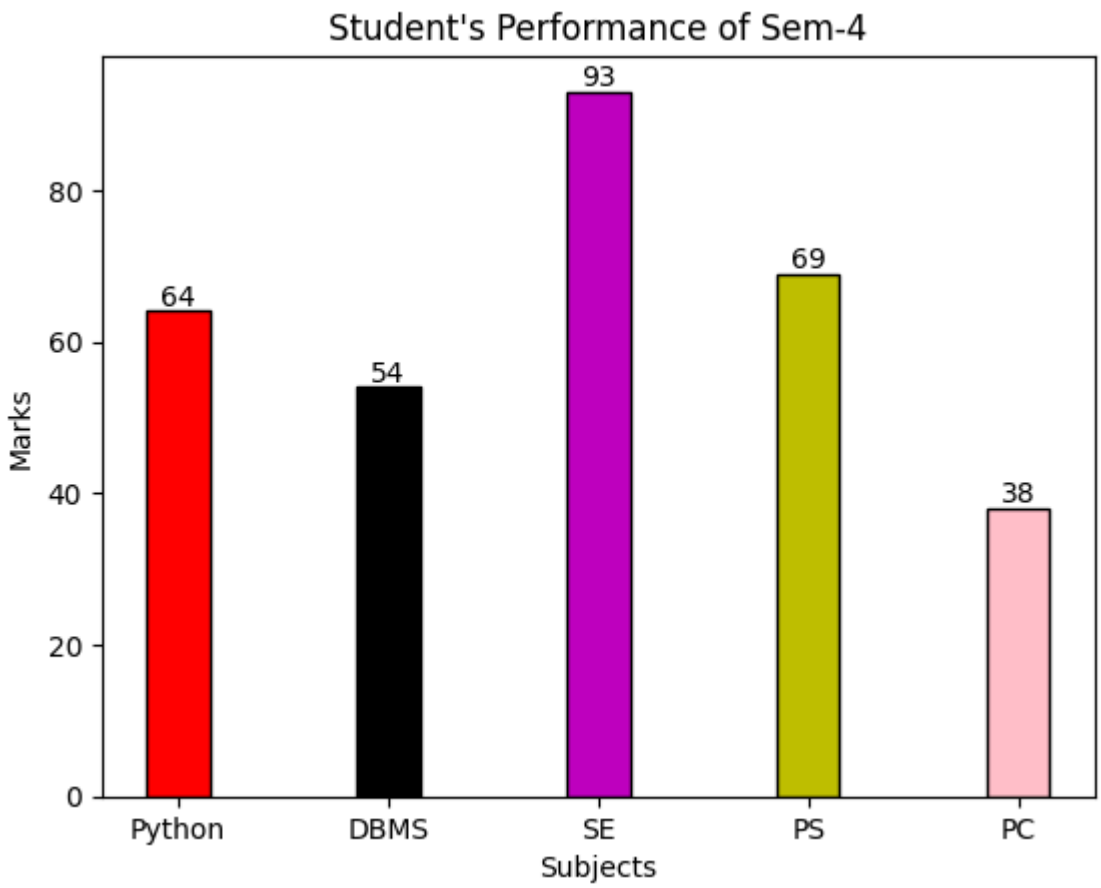
In [42]:
```
marks = [random.randint(35,100) for i in range(5)]
```

In [44]:
```
print(marks)
```

```
[64, 54, 93, 69, 38]
```

Verical Bar Chart

In [46]:
```
bars = plt.bar(subjects, marks, color = ["r","k","m","y","pink"], edgecolor = "k
plt.bar_label(bars)
plt.xlabel("Subjects")
plt.ylabel("Marks")
plt.title("Student's Performance of Sem-4")
plt.show()
```



Horizontal Bar Chart

In [ ]:
```
bars = plt.barh(subjects,marks)
plt.bar_label(bars)
plt.show()
```

Scatter Plot

In [51]:
```
study_hours = range(1,11)
```

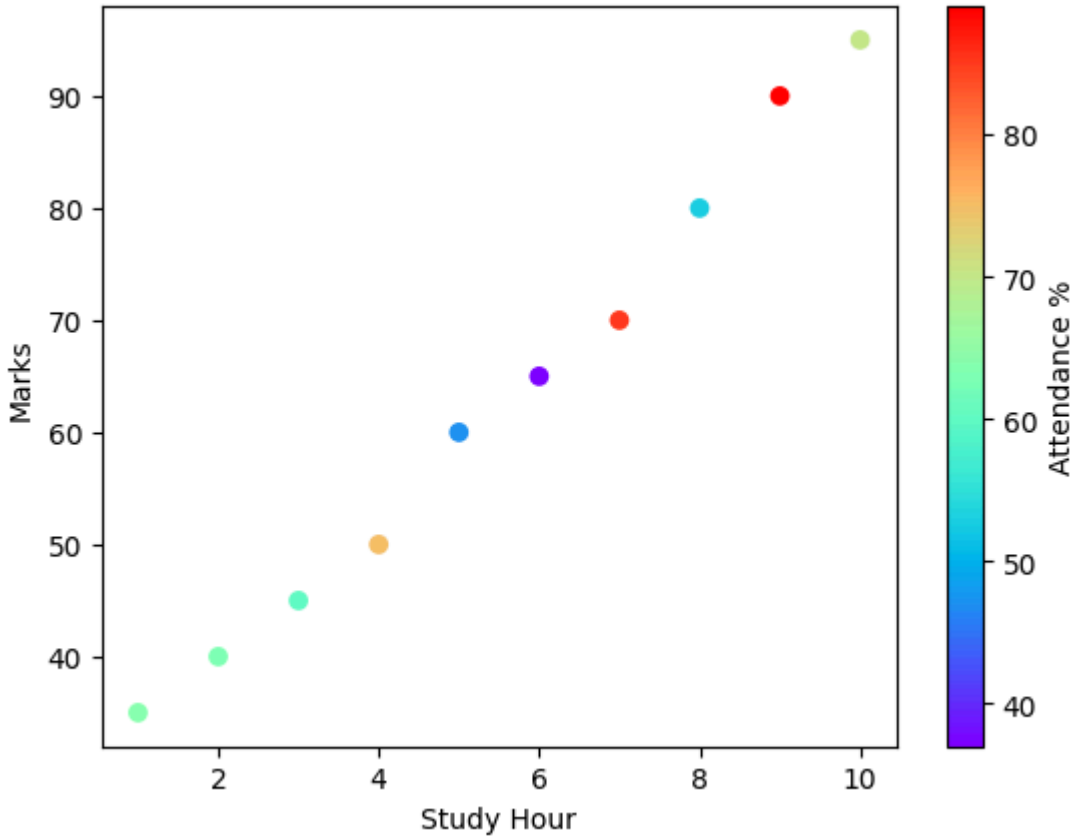
In [53]:
```
marks = [35,40,45,50,60,65,70,80,90,95]
```

In [55]:
```
attendance = [random.randint(35,95) for _ in range(10)]
```

In [57]:
```
print(attendance)
```

```
[64, 63, 60, 75, 47, 37, 85, 53, 89, 70]
```

In [59]:
```
plt.scatter(study_hours,marks,c = attendance, cmap = 'rainbow')
plt.xlabel("Study Hour")
plt.ylabel("Marks")
plt.colorbar(label = "Attendance %")
plt.show()
```



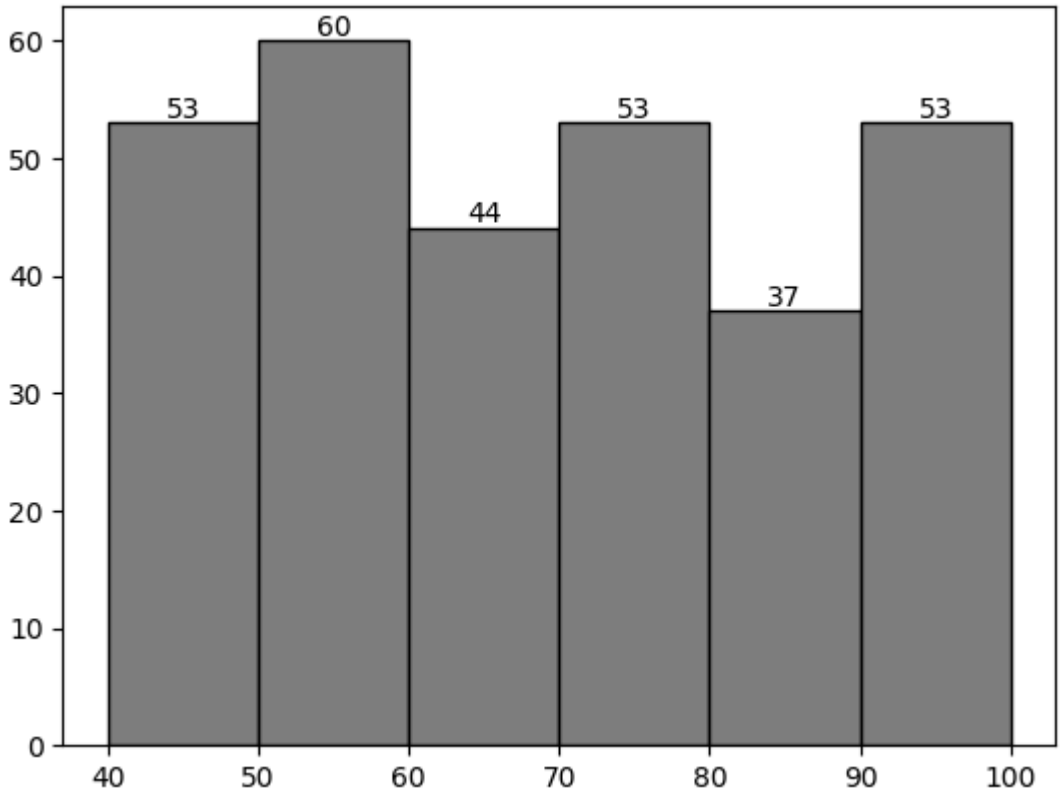
Histogram

In [62]:
```
marks = [random.randint(40,100) for _ in range(300)]
```

In [64]:
```
print(marks)
```

```
[79, 76, 57, 54, 71, 70, 99, 59, 49, 66, 51, 81, 84, 83, 86, 51, 61, 47, 95, 78,
58, 54, 76, 53, 70, 56, 53, 73, 76, 69, 41, 73, 41, 49, 76, 49, 65, 47, 68, 57, 6
1, 61, 43, 50, 90, 49, 59, 95, 97, 53, 66, 97, 92, 76, 59, 75, 69, 47, 83, 99, 7
2, 45, 87, 78, 92, 49, 52, 73, 79, 85, 99, 46, 52, 61, 56, 65, 41, 73, 49, 87, 8
5, 42, 50, 52, 91, 72, 94, 62, 44, 60, 77, 99, 53, 72, 87, 71, 46, 97, 53, 73, 5
3, 75, 90, 44, 90, 74, 95, 50, 92, 52, 95, 92, 52, 59, 65, 92, 98, 45, 43, 51, 8
7, 51, 58, 92, 51, 79, 80, 98, 43, 58, 74, 51, 60, 73, 57, 69, 65, 40, 50, 91, 9
5, 86, 53, 62, 72, 67, 54, 60, 52, 93, 82, 83, 100, 98, 89, 44, 84, 49, 43, 85, 8
3, 62, 87, 90, 41, 57, 64, 63, 66, 82, 71, 41, 89, 42, 74, 76, 65, 62, 86, 47, 4
1, 96, 79, 73, 72, 58, 70, 48, 61, 67, 93, 93, 90, 59, 91, 82, 58, 74, 51, 100, 5
5, 48, 55, 49, 44, 97, 75, 46, 40, 75, 77, 46, 44, 43, 67, 49, 40, 46, 93, 94, 6
6, 65, 90, 48, 86, 71, 53, 55, 58, 49, 47, 100, 86, 66, 50, 62, 72, 86, 89, 66, 9
9, 79, 87, 71, 67, 93, 95, 95, 65, 40, 95, 58, 79, 83, 56, 61, 75, 52, 71, 99, 6
7, 73, 80, 69, 41, 53, 63, 99, 81, 65, 92, 48, 47, 78, 74, 54, 76, 45, 72, 50, 9
9, 57, 80, 55, 58, 89, 99, 82, 55, 79, 91, 61, 84, 89, 51, 46, 94, 65, 45, 63]
```

In [66]:
```
x,y, bars = plt.hist(marks, bins = [40,50,60,70,80,90,100], edgecolor = "k", col
# t1 = plt.hist(marks, bins = [40,50,60,70,80,90,100], edgecolor = "k", color =
plt.bar_label(bars)
#plt.bar_label(t1[2])
plt.show()
```



Boxplot

Boxplot Example

In [ ]:
```
data  = [45, 50, 55, 60, 65, 70, 75, 80, 85, 90]

FIVE NUMBER SUMMARY:

First Quartile (Q1): 55
```

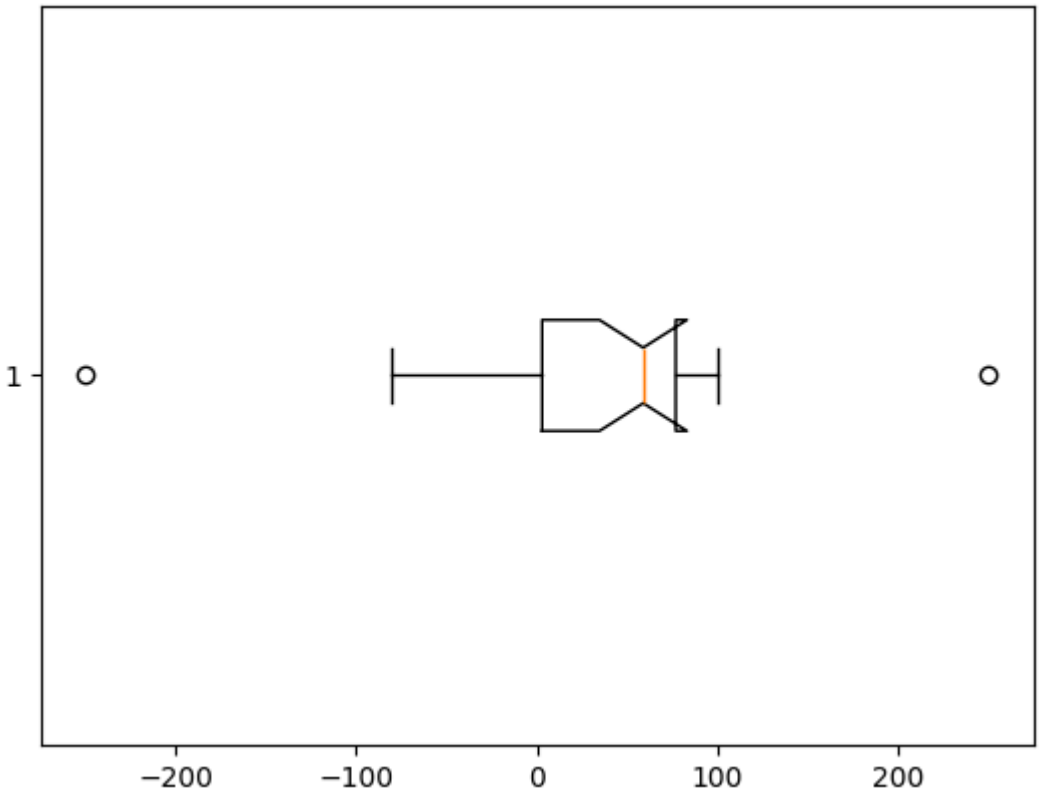
```
Second Quartile (Q2):
Median = 67.5

Third Quartile (Q3): 80

IQR (Inter Quartile Range) (Q3-Q1) = 80-55= 25

Minimum = Q1 - 1.5*IQR = 55 - (1.5*25) = 17.5

Maximum = Q3 + 1.5*IQR = 117.5
```

In [69]:
```
marks = [75,45,70,79,85,82,96,100,65,62,37,59,2,5,4,55,59,-20,-35,250,-70,-80,-2
```

In [71]:
```
plt.boxplot(marks, vert = False, notch=True)
plt.show()
```